Therefore, we can find $2.68 \div 10 = 0.268$;

$2.68 \div 100 = 0.0268$; $2.68 \div 1000 = 0.00268$.

(iii) Division of a decimal number by a whole number

Let us divide a decimal number by a whole number.

Consider the following.

(i) $7.6 \div 2$

$$7.6 \div 2 = \frac{7.6}{2}$$

But $$7.6 = \frac{76}{10}$$

Therefore, $$7.6 \div 2 = \frac{76}{10} \times \frac{1}{2} = \frac{1}{10} \times \frac{76}{2}$$

$$= \frac{1}{10} \times 38 = \frac{38}{10} = 3.8$$

$$\begin{array}{r} 3.8 \\ 2\overline{)\,7.6} \\ 6 \\ \hline 16 \\ 16 \\ \hline 0 \end{array}$$

(ii) Consider $7.26 \div 2$

Now $$7.26 = \frac{726}{100}$$

$$7.26 \div 2 = \frac{726}{100} \times \frac{1}{2} = \frac{1}{100} \times \frac{726}{2}$$

$$= \frac{1}{100} \times 363 = \frac{363}{100} = 3.63$$

$$\begin{array}{r} 3.63 \\ 2\overline{)\,7.26} \\ 6 \\ \hline 12 \\ 12 \\ \hline 6 \\ 6 \\ \hline 0 \end{array}$$

(iii) $9.6 \div 8$.

Now $$9.6 = \frac{96}{10}$$

$$9.6 \div 8 = \frac{96}{10} \times \frac{1}{8}$$

$$= \frac{1}{10} \times \frac{96}{8}$$

$$= \frac{1}{10} \times 12$$

$$= \frac{12}{10} = 1.2$$

$$\begin{array}{r} 1.2 \\ 8\overline{)\,9.6} \\ 8 \\ \hline 16 \\ 16 \\ \hline 0 \end{array}$$

Note

We have to estimate to check that the quotient is reasonably correct. Round 9.6 to 10 and divide by 8. Since 8 goes into 10 once with a small reminder, the answer 1.2 is reasonably correct.

From the above three cases, we observe that we can also divide the number without considering the decimal point first and then the decimal point in the amount may be placed as equivalent to the decimal digits of dividend.

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